

DAVID BECKETT (B.A.SC)

MECHATRONICS ENGINEER



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PROFESSIONAL OVERVIEW

Mechatronics Engineer with 5+ years of experience developing, validating, and manufacturing electro-mechanical products from concept through production. Experienced in mechanical design, PCB integration, hardware testing, supplier coordination, manufacturing support, and continuous product improvement. Proven ability to troubleshoot complex hardware issues, improve manufacturability, and lead cross-functional development projects from prototype to production.

RELEVANT EXPERIENCE

Head of Product Development (Hardware Engineering)

Braze Mobility Inc. | January 2020 - Present

- Led a major product redesign that improved manufacturability, reduced assembly time by approximately 40%, lowered BOM costs by over 20%, and increased overall production efficiency.
- Managed the transition from in-house manufacturing to local contract manufacturing, doubling production capacity while improving production consistency and scalability.
- Designed, validated, and tested robust mounting systems for multiple wheelchair platforms, reducing field failures while improving customer satisfaction.
- Performed root-cause analysis and hardware troubleshooting throughout prototype and production phases, implementing design improvements that increased product reliability and reduced manufacturing issues.
- Managed R&D projects for multiple new product offerings including: data tracking system, speed correction/obstacle avoidance system, a backup camera, new sensor technologies for obstacle detection.

Hardware Engineer

Amina Health | May 2019 – August 2019

- Iterative and collaborative mechanical design of an LFA enclosure using SolidWorks, leveraging rapid-prototyping (FDM, SLA, laser cutting) while focusing on DFM principles for injection moulding.
- Designed, integrated, and validated a battery power management PCB using Altium and SolidWorks, performing electrical and mechanical testing to verify system performance.

Quality Engineer

XYZ International | September 2018 – December 2018

- Designed, prototyped, and validated manufacturing jigs and fixtures to improve production efficiency, repeatability, and product quality.
- Developed production documentation, work instructions, and quality tracking systems to improve manufacturing consistency and support continuous process improvement.

OculAid – Capstone Design Project

University of Waterloo | September 2020 – April 2021

- Design, Prototyping and Testing of a pair of smart glasses to provide obstacle detection and alerting for blind users.
- Designed, fabricated, assembled, and validated custom PCBs, integrating electrical hardware with embedded firmware and mechanical systems.
- Collaborative integration of the electrical design with the firmware and mechanical systems.

EDUCATION

Mechatronics Engineering

University of Waterloo

Dean's Honours List

2016 - 2021

TOOLS

- SolidWorks/Fusion 360
- Altium Designer
- MATLAB/Simulink
- Excel/Power BI
- Arduino

MECHANICAL SKILLS

- FDM, SLA/DLP, SLS 3D Printing
- CNC and Manual Milling
- Laser Cutting
- GD&T
- General Machine Shop Usage

ELECTRICAL SKILLS

- PCB Design and Fabrication
- Through-Hole/SMT Soldering
- Testing and Integration of PCBs

QUALITY/MANUFACTURING

- Manufacturing Support
- Customer Support
- Supplier Coordination
- DFM/DFA
- Root Cause Analysis
- Hardware Validation & Testing

CERTIFICATIONS

- Working at Heights
- Forklift
- Scissor Lift

HOBBIES/INTERESTS

- Ice Hockey
- Volleyball
- 3D Printing
- Camping
- Golf
- Formula 1